

Module :Window Networking Service

1. Discuss the role of Windows Firewall in Windows Server and how to configure it

Role of Windows Firewall

Windows Firewall protects the server from unauthorized access by filtering incoming and outgoing network traffic.

Functions:

- Blocks unauthorized connections
- Allows trusted apps/services
- Controls inbound and outbound traffic
- Protects against malware/network attacks
- Logs network activity

Configuration Steps

Method 1: GUI

1. Open Server Manager
2. Go to Tools → Windows Defender Firewall with Advanced Security
3. Select:
 - Inbound Rules
 - Outbound Rules
4. Click New Rule
5. Select rule type:
 - Program
 - Port
 - Predefined
 - Custom
6. Allow/Block connection
7. Choose profile:
 - Domain
 - Private
 - Public
8. Save rule

Method 2: CMD

netsh advfirewall set allprofiles state on

Allow port 80:

netsh advfirewall firewall add rule name="HTTP" dir=in action=allow protocol=TCP localport=80

2. What is NAT in Windows Server and how do you configure it?

NAT (Network Address Translation)

NAT converts private IP addresses into public IP addresses so devices can access the internet.

Uses

- Saves public IP addresses
- Hides internal network
- Improves security

Configuration Steps

1. Open Server Manager
2. Add Roles:
Remote Access
3. Select:
 - Routing
4. Install role
5. Open Routing and Remote Access
6. Right-click server → Configure and Enable RRAS
7. Select:
NAT
8. Select public internet interface
9. Enable private interface
10. Finish

Verify:

ipconfig

ping google.com

3. Explain DHCP and how to configure it in Windows Server 2016

DHCP Full Form

Dynamic Host Configuration Protocol

Purpose

Automatically assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server

Without DHCP, IP must be set manually.

Configuration Steps

Install DHCP Role

1. Open Server Manager
2. Add Roles and Features
3. Select DHCP Server
4. Install

Post Deployment

Click:

Complete DHCP Configuration

Authorize DHCP server in Active Directory.

Create Scope

1. Open DHCP
2. Expand server
3. Right-click IPv4
4. Select New Scope

Enter:

- Name: LAN Scope
- Start IP: 192.168.1.10
- End IP: 192.168.1.200
- Subnet Mask: 255.255.255.0

Add exclusions if needed.

Set:

- Gateway: 192.168.1.1
- DNS: 8.8.8.8

Activate scope.

Verify

Client PC:

ipconfig /release

ipconfig /renew

ipconfig /all

4. Describe the process of configuring DNS in Windows Server

DNS Full Form

Domain Name System

Converts domain names into IP addresses.

Example:

google.com → 142.x.x.x

Configuration Steps

Install DNS

1. Server Manager
2. Add Roles
3. Select DNS Server
4. Install

Create Forward Lookup Zone

1. Open DNS Manager
2. Expand server
3. Right-click Forward Lookup Zones
4. New Zone
5. Primary Zone
6. Enter domain:

company.local

Add Host Record

Right-click zone → New Host

Example:

Name:

www

IP:

192.168.1.10

Now:

www.company.local

Verify

nslookup www.company.local

ping www.company.local

5. What is Server Manager and how is it used?

Definition

Server Manager is the main management console in Windows Server.

It helps manage:

- Roles
- Features
- Performance
- Storage
- Services
- Remote servers

Uses

- Add/remove roles
- Monitor alerts
- Configure DNS/DHCP/AD
- Manage local & remote servers
- Check event logs

Open Server Manager

servermanager

Or:

Start Menu → Server Manager

6. Discuss the role of Remote Desktop Services (RDS) and how to configure it

Role of RDS

Allows users to remotely access Windows Server desktop/applications.

Used for:

- Remote administration
- Centralized application hosting
- Multi-user access

Benefits

- Work from anywhere
- Central management
- Saves hardware cost

Configuration Steps

Enable Remote Desktop

1. Server Manager
2. Local Server
3. Click Remote Desktop
4. Select:

Allow remote connections

Apply

Add RDS Role

1. Add Roles and Features
2. Select:

Remote Desktop Services

Install:

- RD Session Host
- RD Licensing
- RD Connection Broker

Restart server

Allow Firewall Rule

```
netsh advfirewall firewall set rule group="remote desktop" new enable=Yes
```

Connect from Client

Press:

Win + R

mstsc

Enter server IP:

192.168.1.10

Login with admin credentials.